Control Theory

Chap 1: Introduction

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Rm 403, Bld. Aerospace

October 15, 2026



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Course Information

About Our Class

Course Hour:	32 Teaching Hours
Where:	Rm 335, Bld. 3, Zhongguancun Campus
When:	Tuesday, 11-13 Class, Week 4-9,
	Thursday, 11-13 Class, Week 4-8.
Office Hour:	13:00-18:00,Wednesday and Friday,
	Rm403, Bld. Aerospace.
	(Better Appointment)
Contact Us:	gzhai@bit.edu.cn, & sunyy@bit.edu.cn,
	136-5102-2718(Wechat)

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Course Information

Objective:

- Not Math but for Engineering Application;
- Not for Examination but for Research;
- Review and Expand what undergraduates studied;
- Share ideas each other;
- Known more about automation and helpful for your researches.

What we talk:

- Model: Differential model, Laplace Model, State-space model,...
- Analysis: Transient and Steady-state characteristics, Stability,...
- Design: Observer, Controller, Optimization...
- Application: Aerospace, Robot, Rocket, Electrical Motor, Thermal System, Electronics, Public Information, Virus Control, Criminal, Human Resource, ...

Course Information

Exam

No Written Exam

Class Participation:	25%, Attend class and participate constructively
Homework:	25%, Software simulations and Some Exercises
Final Project:	25%, Academic Presentation, About 8-10 $\min$
	25%, Study Report, Academic Paper Form, *Electronic Version, PDF or Word*.
	English (CHN Stu.).
	English or Chinese (International Stu.).
	Better related with your research and control.

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Course Information

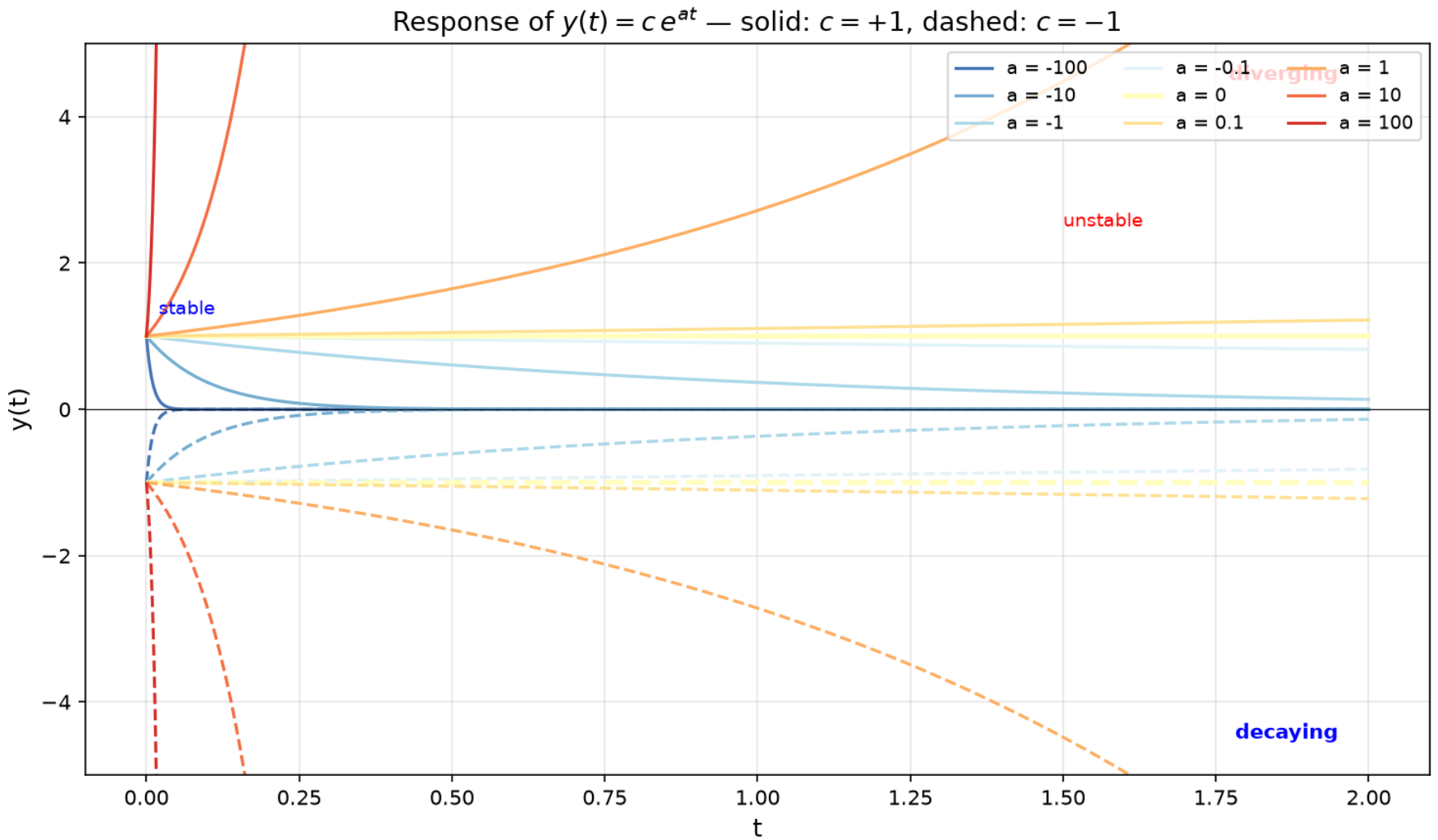
Necessary Knowledge

- Laplace Transformation $\mathfrak{L}[f(t)] = F(s)$ and Inverse Laplace Transformation $\mathfrak{L}^{-1}[F(s)] = f(t)$;
- Linear algebra theory:vector, matrix, rank, determinant, eigenvalue, transpose, reverse, etc.;
- Differential theory: Integrator $\int f(x)dx$, differential $\frac{d(f(t))}{dt}$;
- Exponents and Logarithms: matrix form e^A, scalar form e^a, $\log_a b$ and $\ln(a)$;
- Time domain and Frequency domain.

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Question

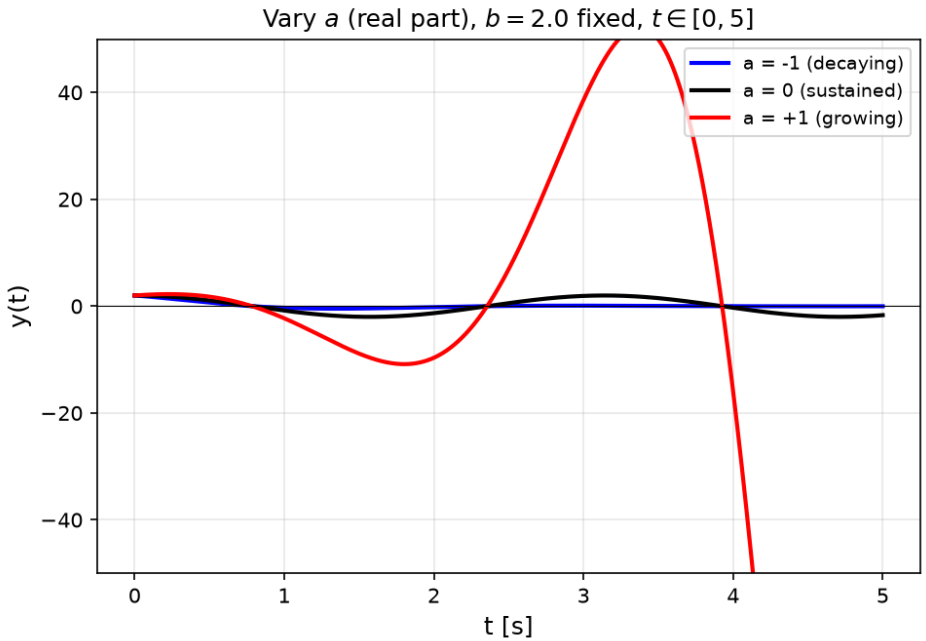
Given $y(t) = c\,e^{at}$ with $c = \pm 1$, how does the response change for $a \in \{-100, -10, -1, -0.1, 0, 0.1, 1, 10, 100\}$?

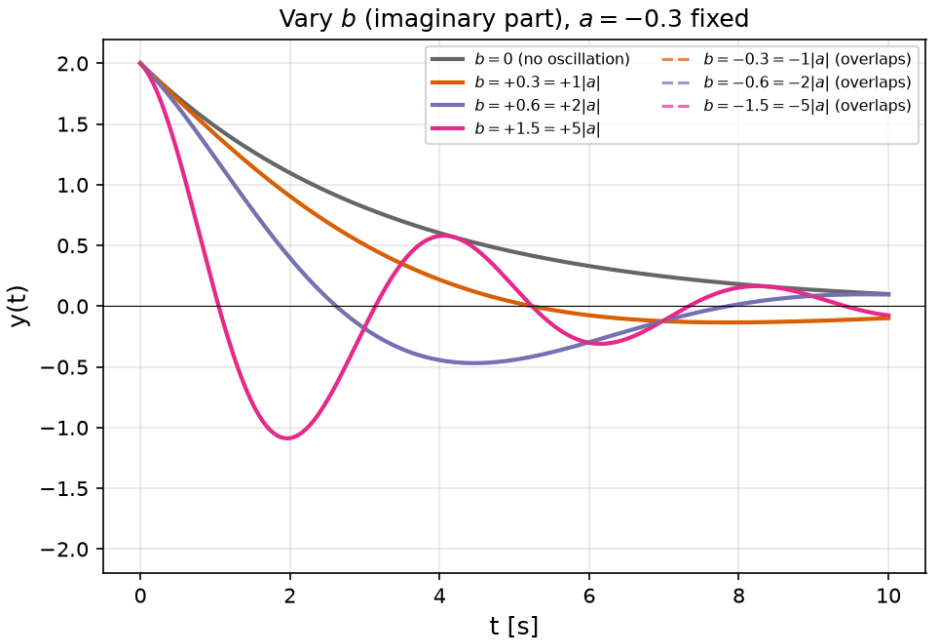


Question

Given $y(t) = c_1 e^{(a+bj)t} + c_2 e^{(a-bj)t}$ with $c_1 = c_2 1$, how does the response change with a (real part) and b (imaginary part)?

$c_1 = c_2 = 1$: $y(t) = e^{(a+bj)t} + e^{(a-bj)t} = 2e^{at}\cos(bt)$





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Reference

- Katsuhiko Ogata, Modern Control Engineering, 5th Edition
- Burzo E. PID control: new identification and design methods[M]. Springer, 2010.
- 刘金琨，先进PID控制Matlab仿真第4版，北京:电子工业出版社，2016年6 月
- Liu J. Advanced PID control and MATLAB simulation[J]. Beijing: Publishing House of electronics industry, 2004, 9.
- Yiyong Sun, Guang Zhai, and Yao Zhang. 控制理论与工程应用(英)，Control Theory and Engineering Application – Textbook for Postgraduates. Not published, 北京理工大学出版社，2026,7

Code: `https://github.com/Syybit2024/Control_Theory_for_PostGraduates`

APP：Experienced Controls, By Quanser

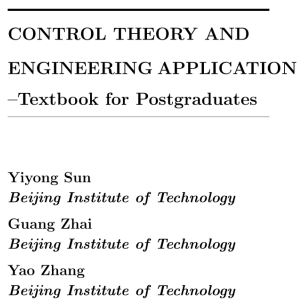




Matlab, 2020+



Text Book



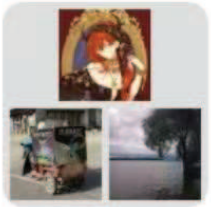
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Course Information

Class Plan

Chap 1 Introduction:	2 Hours, Week 4
Chap 2 Dynamics and Model:	4 Hours, Week 4
Chap 3 System Analysis:	6 Hours, Week 5
Chap 4 PID :	3 Hours, Week 6
Chap 5 State-space System:	6 Hours, Week 6, 7
$*$ Chap 6 Optimal Control :	3 Hours, Week 7
$*$ Chap 7 Kalman Filter :	3 Hours, Week 8
Student Presentation:	6 Hours, Week 8 and 9

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群聊：Control theory 2026

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Chap 1: Introduction

Objectives:

1. The global key historical developments in control theory.
2. The evolution of science and technology to control, especially on aerospace engineering and unmanned autonomation.
3. Recognize fundamental modes of automatic operation.
4. Grasp foundational principles governing the analysis of systems.
5. Master the standard design methodology for feedback controllers.



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Chap 1: Introduction

1. History of Automatic Control
2. Fundamentals of Control
3. Classification of Automatic Control Mode
4. Basic Requirements of Automatic Control System
5. Design Process of Control

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1 History of Automatic Control

2 Fundamentals of Control

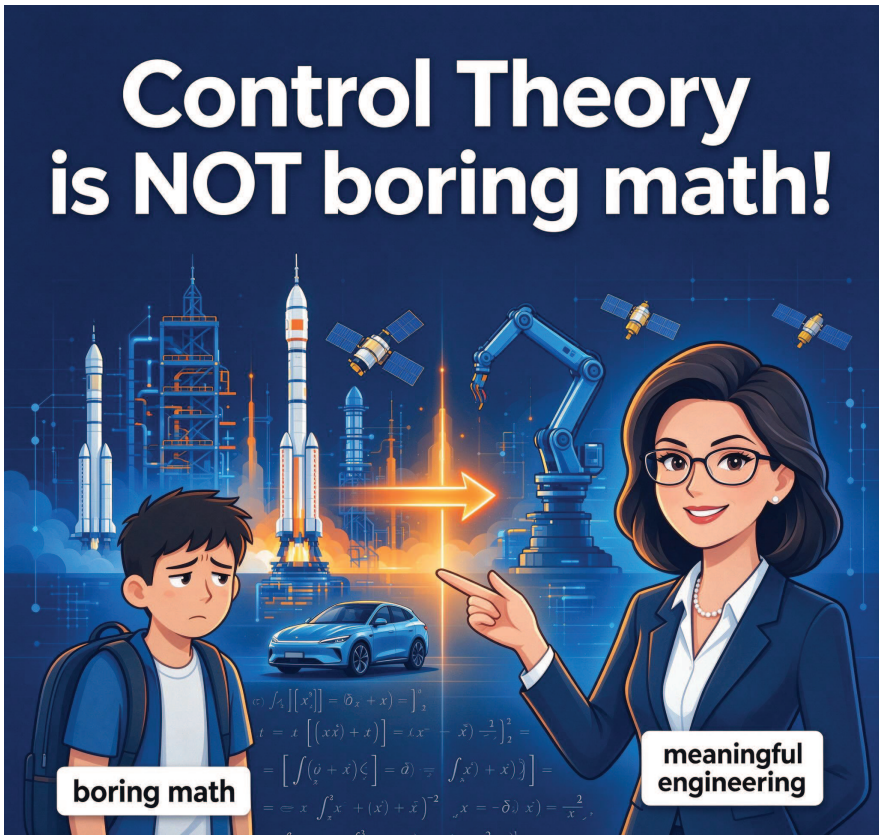
3 Classification of Automatic Control Mode

4 Basic Requirements of Automatic Control System

5 Design Process of Control

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What's Control?





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Definition

Control (Automation, Automatic Control, Cybernetics...)
A control system is an interconnection of components forming a system configuration that will provide a desired system response.
Automatic control system: A control system without people involved.

History

Classical control: 1868–1960	the ancient: China, Greece, Rome, Arabia ... the industrial revolution the beginnings of telecommunication the appearance of computers the beginning of space research
Modern control: 1960–1980	State-space-based optimal control and adaptive control
Robust and Nonlinear control: 1980–2000	H_∞ control, robust control, nonlinear c
Intelligent and Learning control: 2000–present	Learning, computer-aided control, $\cdots$

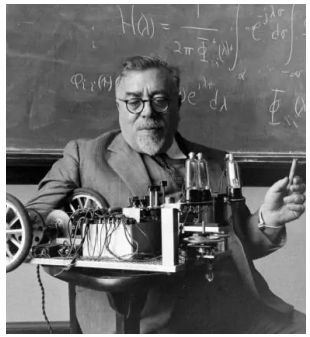
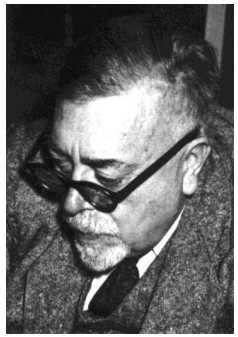
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Those Memorable People

James Watt
Edward John Routh
Adolf Hurwitz
Norbert Wiener
A.M. Lyapunov
Hendrik Wade Bode
Harry Nyquist
Lev Pontryagin
Richard Ernest Bellman
Walter Richard Evans (Root Loci)
Rudolf E. Kalman
Vadim Ivanovich Utkin
Hsue-shen Tsien
Lotfi Aliasker Zadeh
. . .


Edward John Routh
1831–1907

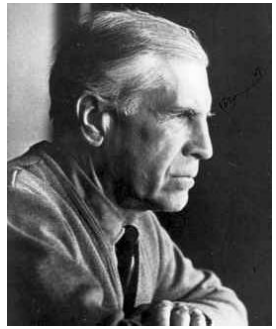

Adolf Hurwitz
1859-1919



Norbert Wiener
1894—1964

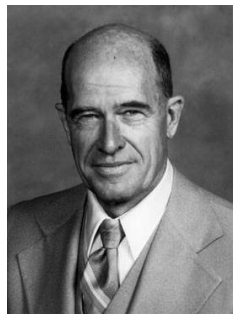

A.M. Lyapunov
1857-1918


Hendrik Wade Bode
1905-1982

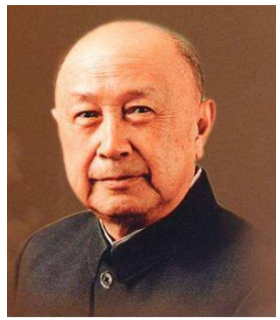

Harry Nyquist
1889-1976


Lev Pontryagin
1908-1988


Richard Ernest Bellman
1920 – 1984


Walter Richard Evans
1920-1999


Rudolf. E. Kalman
1930-2016


Hsue-shen Tsien
1911-2009

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1 History of Automatic Control

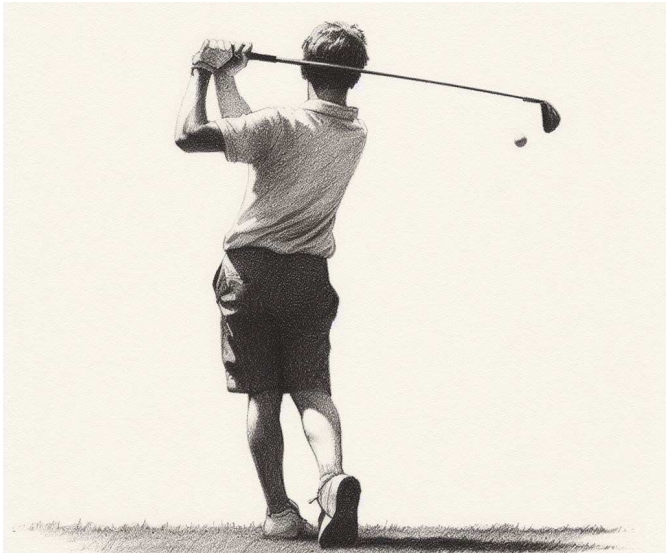
2 Fundamentals of Control
- Open-loop and Closed-loop
- Feedback Control System
- Compound Loops

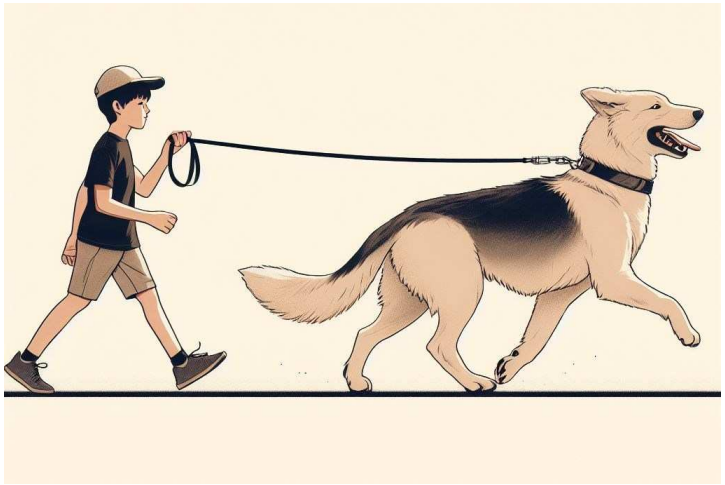
3 Classification of Automatic Control Mode

4 Basic Requirements of Automatic Control System

5 Design Process of Control

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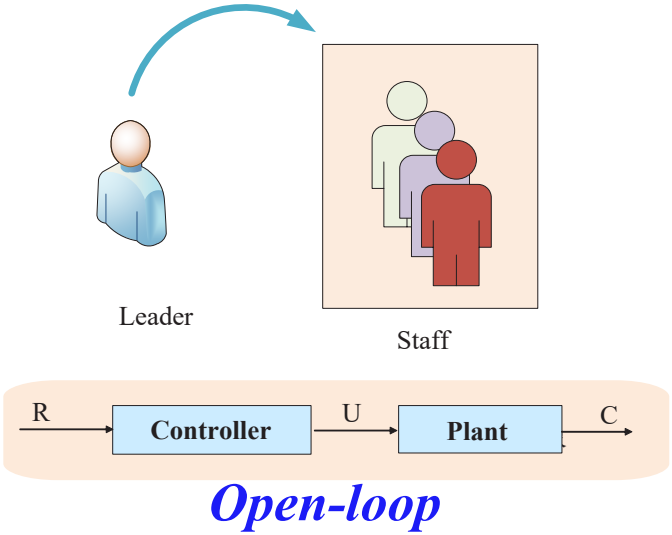




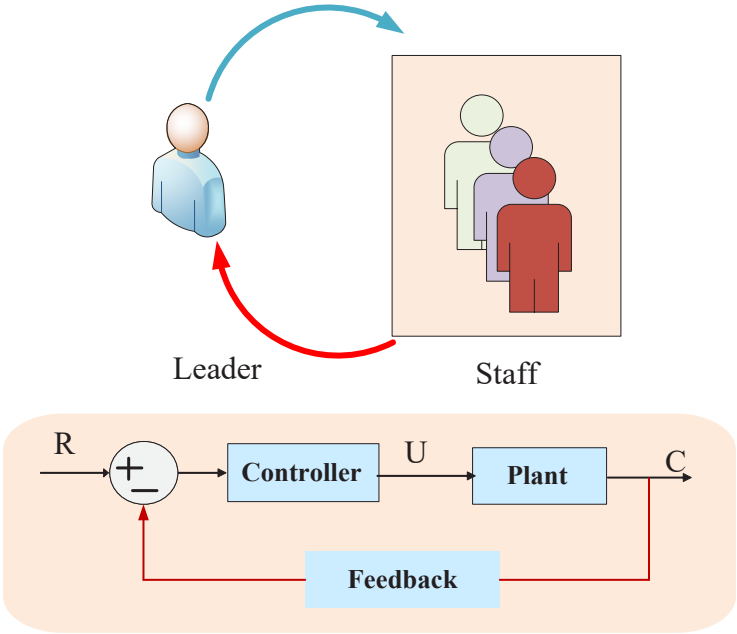




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Open-loop



Closed-loop

Advantages: Simple construction, Ease of maintenance, Less expenses, No measurement,...
Disadvantages: Poor reliability, flexibility, accuracy, and against Disruption...

Advantages: Recalibration, System sensitivity may be small, Less affected by noise,...
Disadvantages: Complex and Expensive, Feedback may lead to oscillation, Reduces the overall system gain, More for Stability,...

1 History of Automatic Control

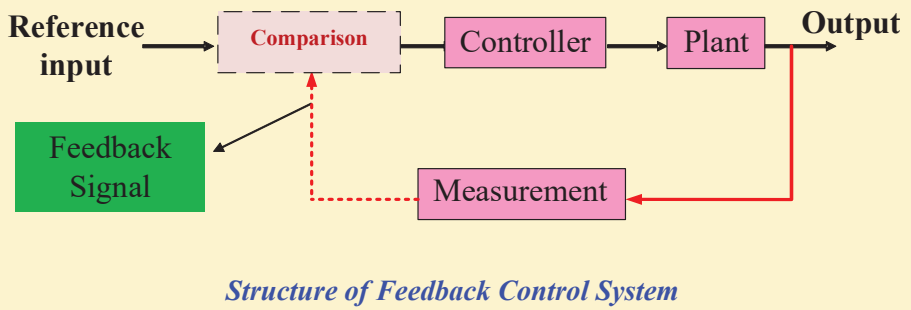
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Structure of Feedback Control System

How to Feedback

Measurement or Sensor,
Comparison Input and output
Compute correction
Apply to system or Actor

What's Feedback Control

System correcting itself using output measurements.

What Feedback Can do?

Rocket, Satellite, Robot, Automatic Driving, Business, Criminal Control, Human Resource,Virus, Rumours, Electronics， Electronic Motor, Thermal and Air Adjustment, ...

Why Feedback

Insensitive to external disturbances,
Still properly work with uncertainties,
To achieve accuracy, stability.

Feedback changes the way humans think.

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1 History of Automatic Control

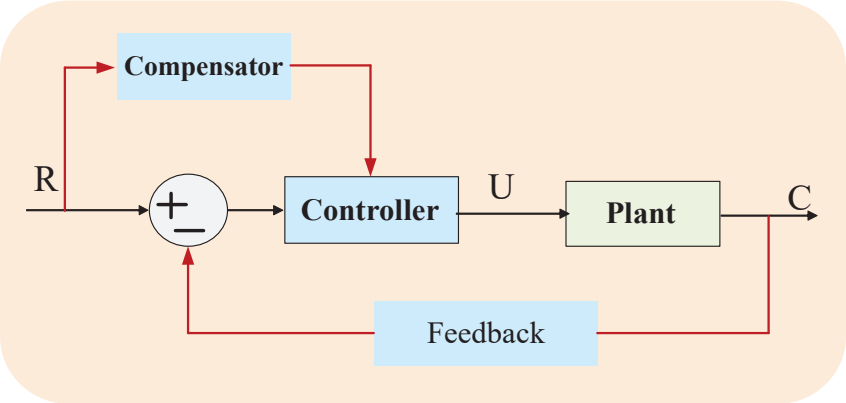
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3 Classification of Automatic Control Mode

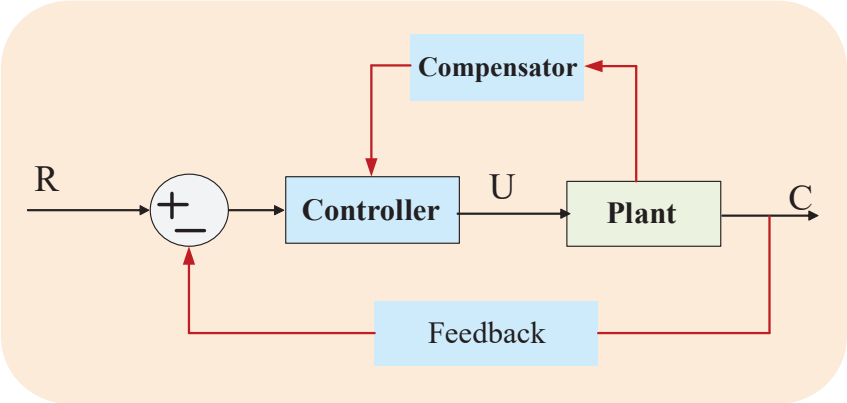
4 Basic Requirements of Automatic Control System

5 Design Process of Control

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Closed-loop + Input Compensator



Closed-loop + Disturbance Compensator

- Combination of open-loop and closed-loop structure;
- Connection of serial and parallel.

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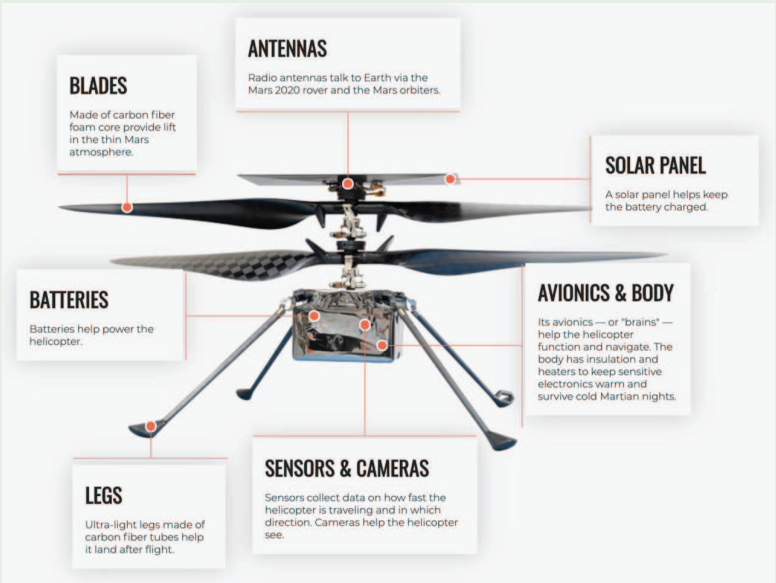
Example (Bicycle and Unmanned Bicycle)





	Rider Bicycle	Unmanned Bicycle
Sensors	Eyes, Body gesture	Cameras, Radars, IMU, GPS
Actuators	Legs and Hands	Motors
Controllers	Brain	Computer
Feedback aims	Balance, Tracking, Speed, Human-body	Balance, Speed

Example (Unmanned Helicopter)

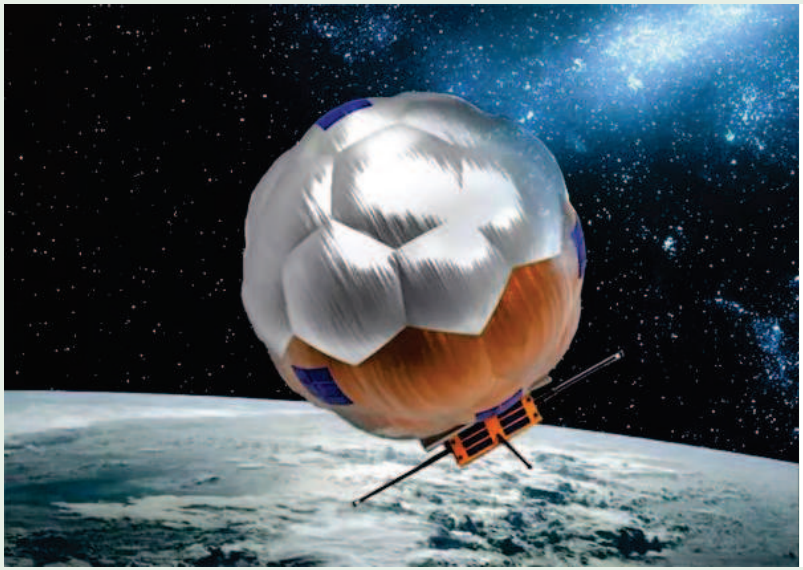


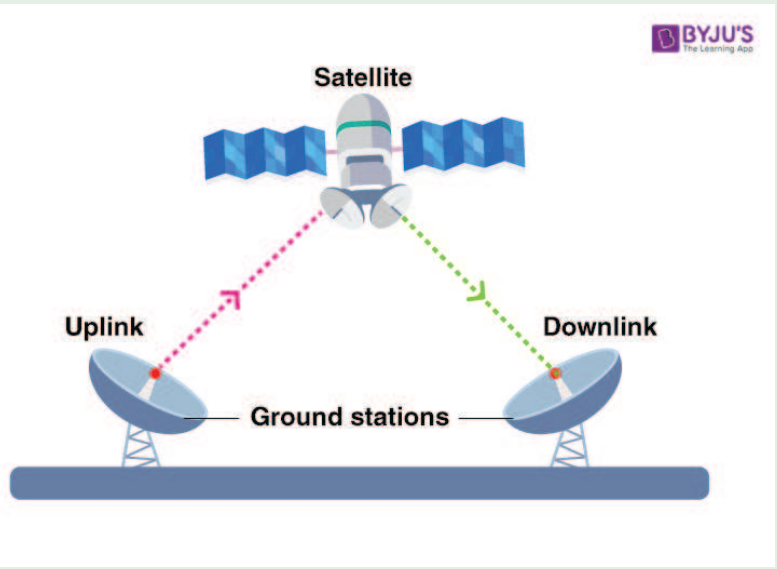


	Ingenuity (Ginny)
Sensors	Cameras(RGB+Black-white), Rotation-sensor
Actuators	Rotary Motors , Pitch link
Controllers	MCU
Feedback aims	Altitude, Movement(Horizon, and Vertical)

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Example (Satellite: BIT No.1)

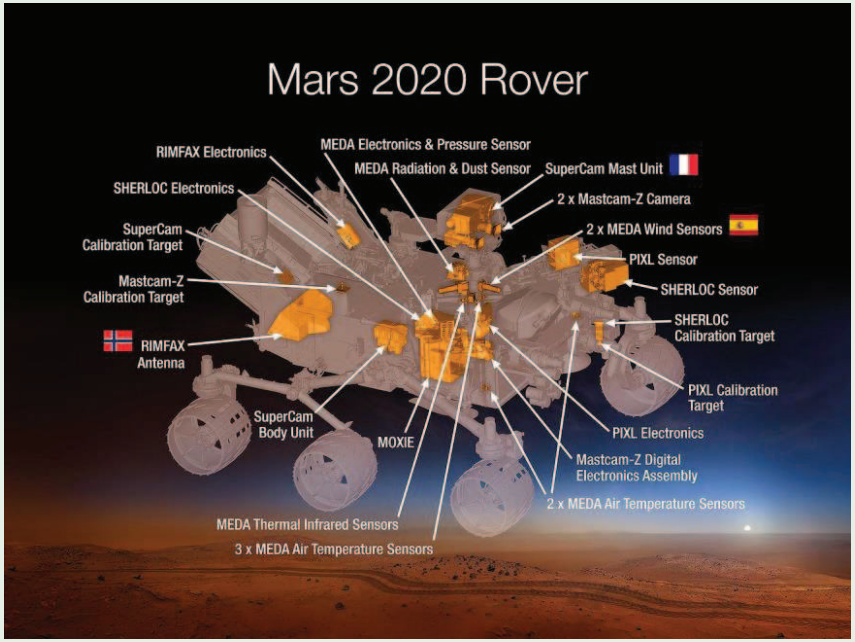


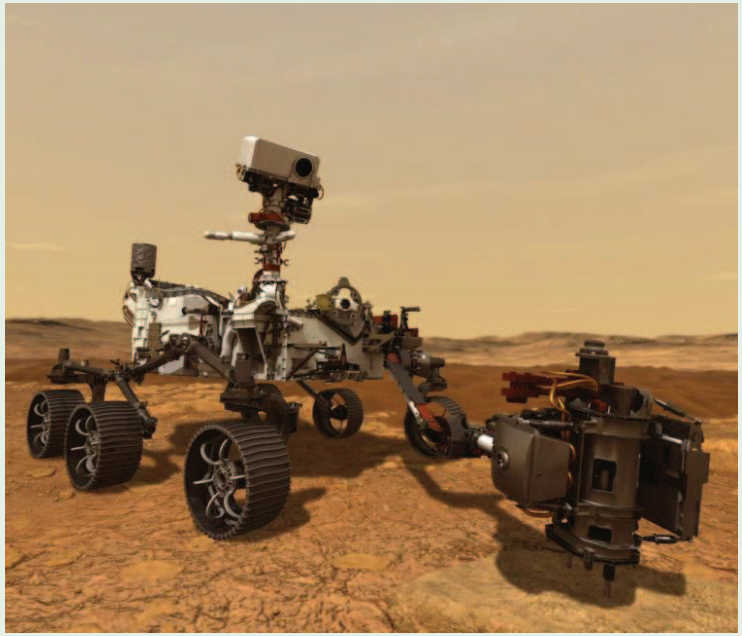


	Satellite
Sensors	Ground Radar, Visible-light and Infrared cameras, IMU
Actuators	Fuel Injector, Control-moment-gyro(CMG)
Controllers	Aerospace Computer
Feedback aims	Altitude, Track

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Example (Unmanned Rover)



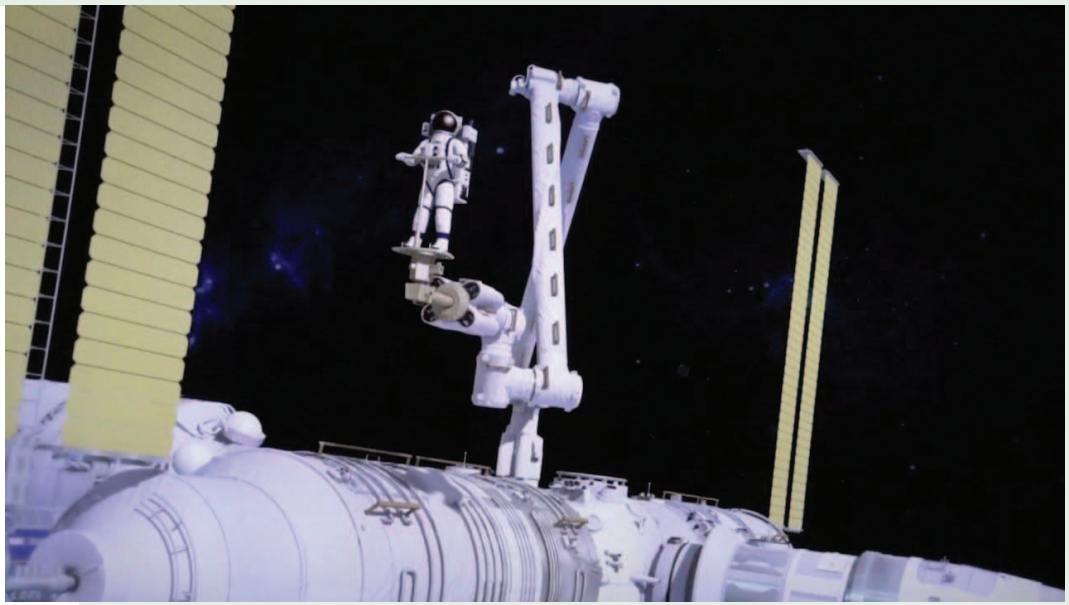


	Perseverance Unmanned Rover
Sensors	Cameras, Wireless Location, Rotation Sensor,...
Actuators	Motors, Manipulator arms,...
Controllers	Aerospace Computer
Feedback aims	Altitude, Trajectory tracking

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Example (Manipulator and Space Arm)





	Manipulator
Sensors	Encoder,...
Actuators	Motors, ...
Controllers	Embedded-computer
Feedback aims	Axis-rotation, Trajectory tracking

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1 History of Automatic Control

2 Fundamentals of Control

3 Classification of Automatic Control Mode
- Classification
- Classical and Modern Control
- Linear and Nonlinear Systems

4 Basic Requirements of Automatic Control System

5 Design Process of Control

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Classification

Appearance Time	Classical, Modern, Robust and Nonlinear, Intelligent and Learning ...
Control Mode	Open-loop, closed-loop and compound Feedback, Feed-forward and compound
System Performance	Linear v.s. Nonlinear systems, Continuous, Discrete and Hybrid systems, Time-invariant v.s. Time-varying systems, Deterministic v.s. Uncertain systems SISO v.s. MIMO
Plant Type	Mechanical, electrical, Electromechanical, hydraulic , pneumatic, biological, virus, thermal, population, economical systems
Model Dependence	Model-dependent v.s. Model-free
Control Method	PID, State-feedback, Fuzzy, Sliding mode, Neural-network, Optimal Control, Robust...

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Classical v.s. Modern Control

	Classical	Modern
Object	LTI, SISO	Time-varying, Nonlinear, MIMO
Domains	Time and Frequency domains	Time domain
Math Tools	Differential equation Laplace transformation	Linear algebra Matrix theory
Analysis Methods	Time-domain and Frequency domain, Root-locus	State-space
Description	Input-output	Inside the plant
Design Methods	PID	Pole placement
Limitations	Shortages on complex, Multi-variable, Time-varying and Nonlinear systems	Complex computation



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Principle of Superposition

A system, defined by the function $f(x)$, is linear if

$$\begin{aligned} f(x_1) &= y_1, f(x_2) = y_2 \\ f(x_1 + x_2) &= y_1 + y_2 \end{aligned}$$

or: Additivity and Homogeneity

A system (function) is linear if it satisfies

$$\begin{aligned} f(x_1 + x_2) &= f(x_1) + f(x_2) \quad \textbf{Additivity} \\ f(cx_1) &= cf(x_1) \quad \textbf{Homogeneity} \end{aligned}$$

Example

Determine whether the functions are linear or not

$$\begin{aligned} f_1(x) &= ax(t) + b, \\ f_2(x) &= ax^2(t) + bx(t) + c \\ f_3(x) &= a\dot{x}(t) + b\int_0^t x(\tau)d\tau + cx(t) + d \\ f_4(x) &= a\sin(x(t)) + b\dot{x}(t) \end{aligned}$$

LTI v.s. LTV

Example

Determine whether the linear functions are time-variant or time-invariant

$$\begin{aligned} f_1(x,t) &= a_1(t)x(t) \\ f_2(x,t) &= a_2(t)\dot{x}(t) + b_2(t)\int_0^t x(\tau)d\tau \\ &\quad + c_2(t)x(t) \end{aligned}$$

Example

Electrical components: Resister R and $R(t)$.
Rocket fuel consumption: M and $M(t)$.

Plane drag caused by Air: $F = \frac{1}{2}C(t)Sv^2$

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1 History of Automatic Control

2 Fundamentals of Control

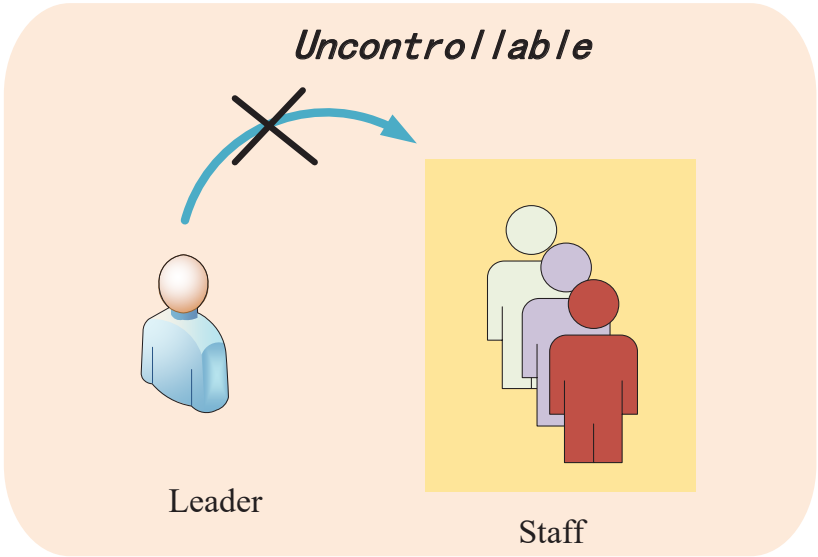
3 Classification of Automatic Control Mode

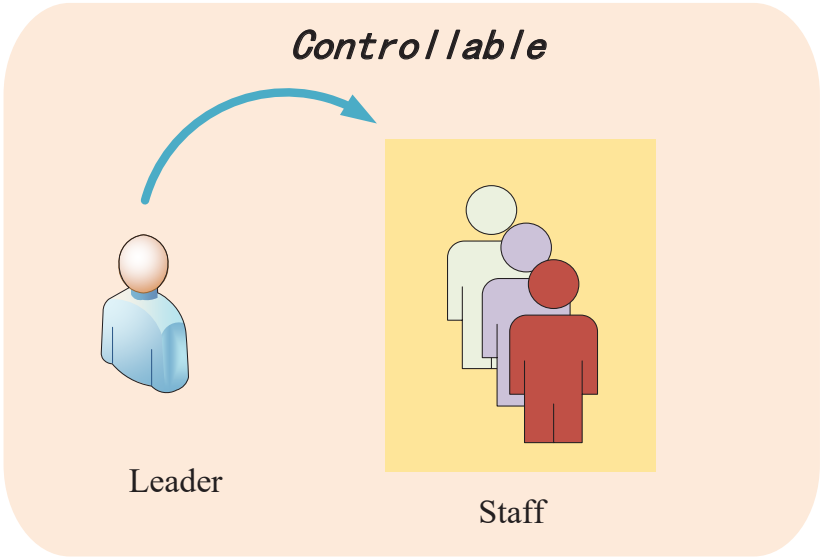
4 Basic Requirements of Automatic Control System
- Controllability and Observability
- Stability, Rapidity, Smoothness, Accuracy
- Typical External Functions

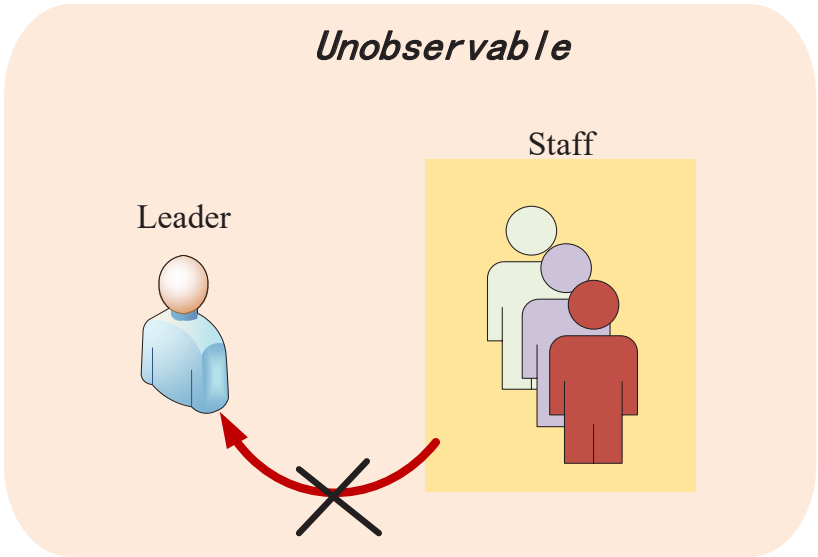
5 Design Process of Control

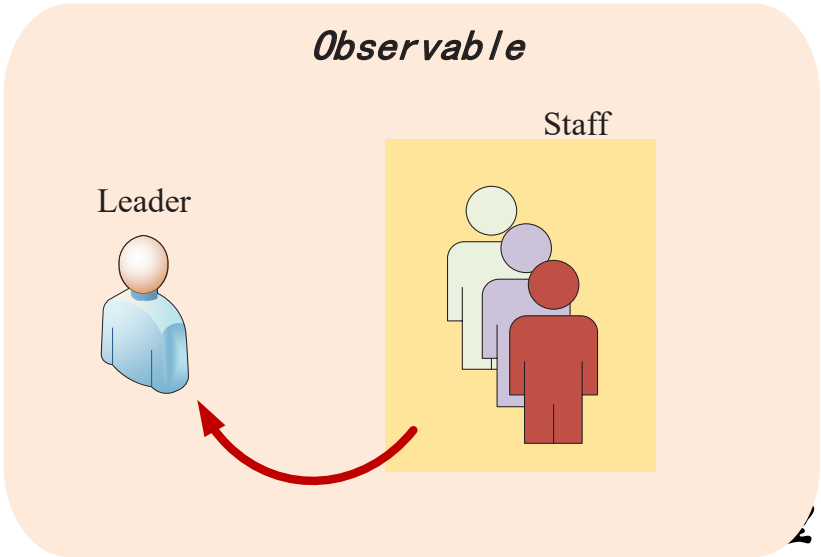


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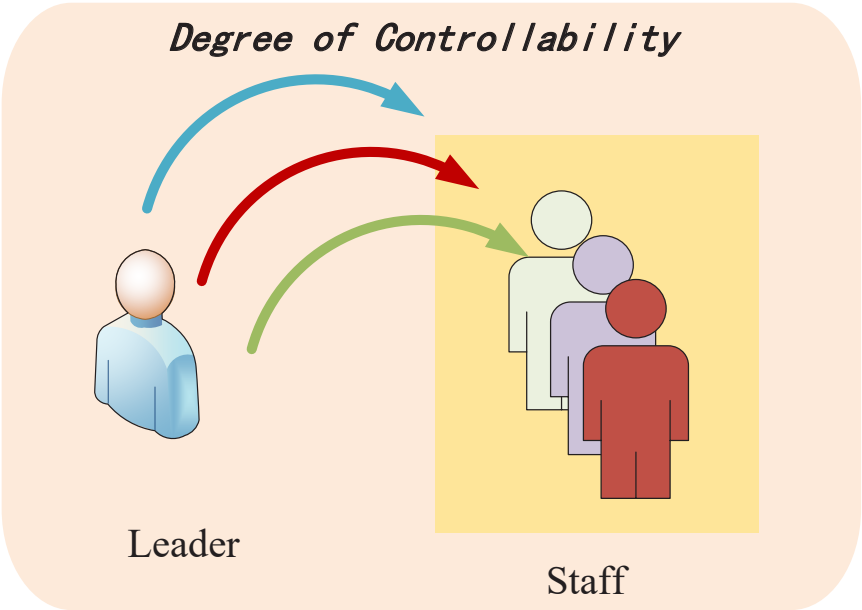

Uncontrollable
Leader
Staff

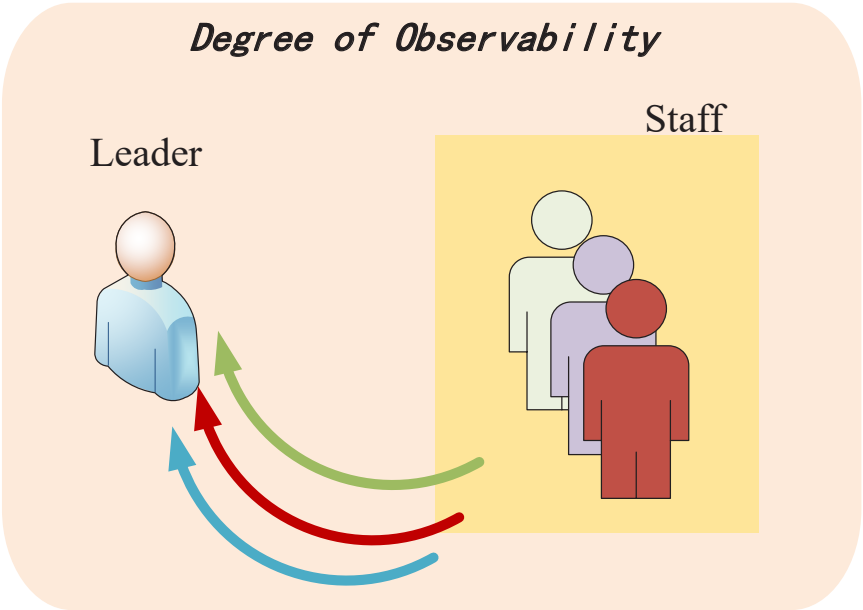

Controllable
Leader
Staff


Unobservable
Staff
Leader


Observable
Staff
Leader

The degree of controller can affect the system plant.
The degree of system plant can be measured.





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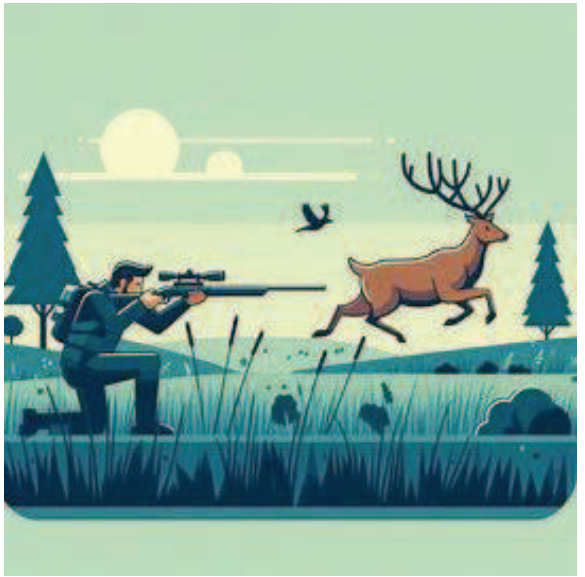
Degree of Controllability - Easier and simpler hunting



Stone and Stick



Bow and Arrow



Rifle and Gun

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Degree of Observability - More accurate diagnosis




Excrement and Urin



Stethophone




CT and Nuclear Magnetic Resonance



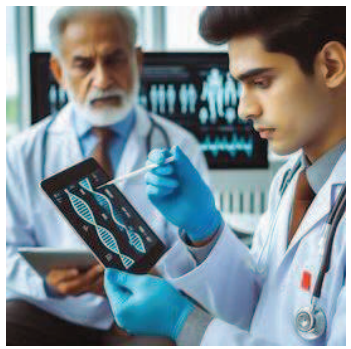
Feel the Wrist Pulse



Tongue Coat



Hamnatodynamometer

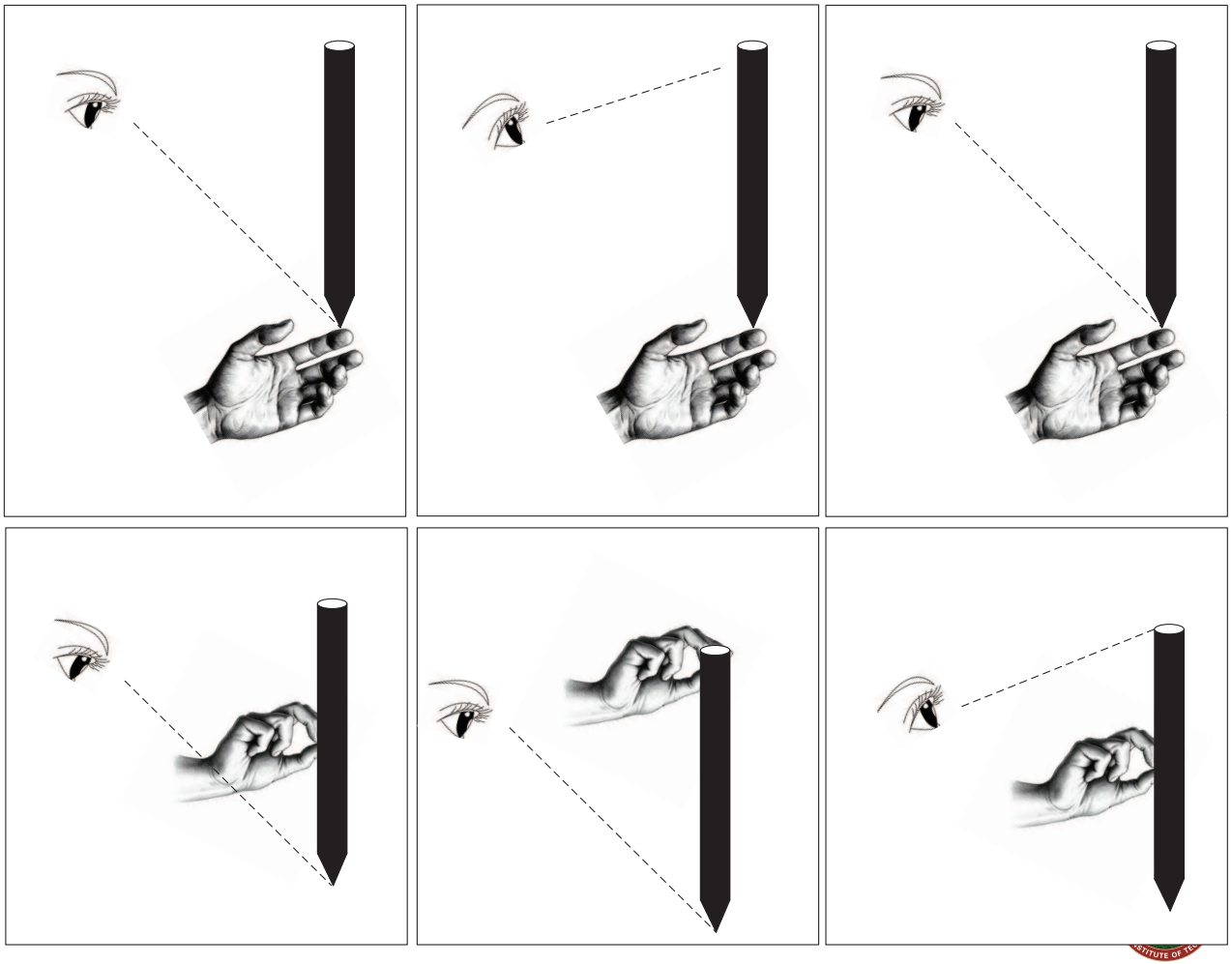



DNA and X-ray Check

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Degrees of Controllability and Observability

How much controller can affect the system plant.
How much system can be monitored and traced via measurement...



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1 History of Automatic Control

2 Fundamentals of Control

3 Classification of Automatic Control Mode

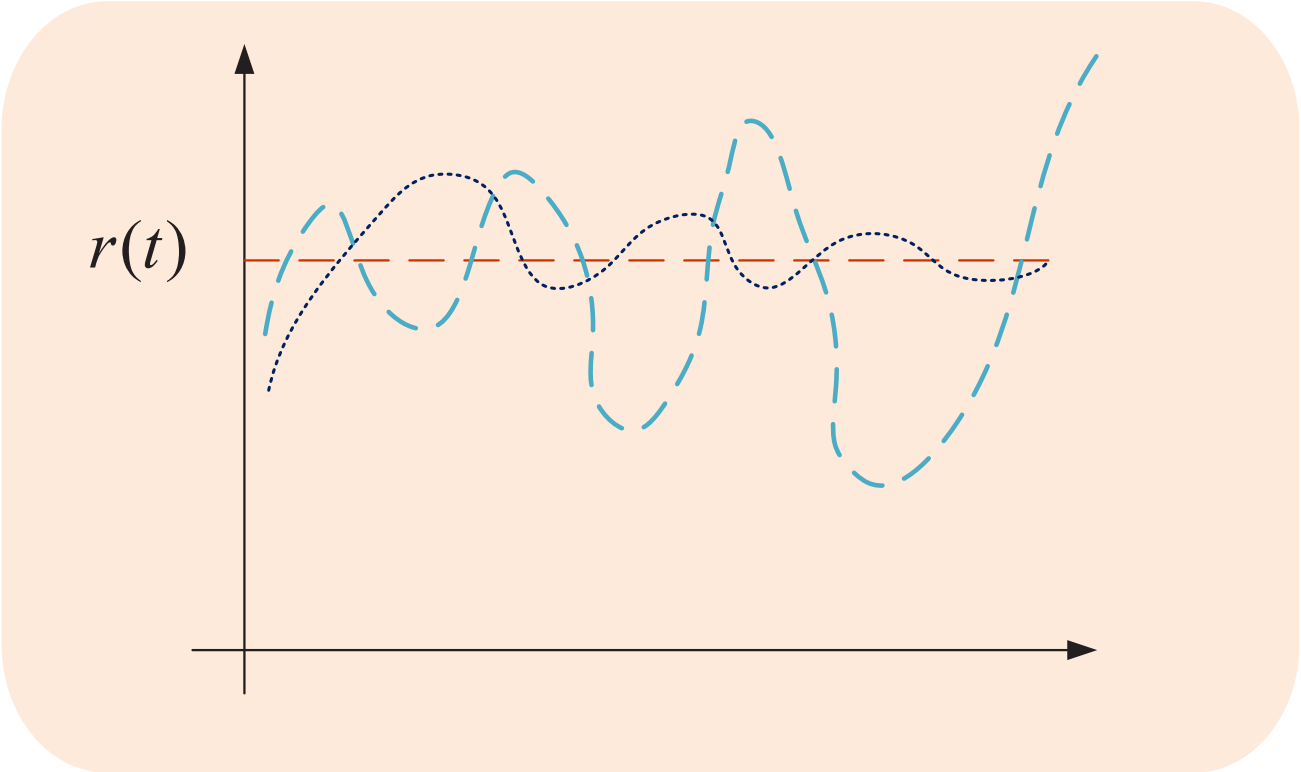
4 Basic Requirements of Automatic Control System
- Controllability and Observability
- Stability, Rapidity, Smoothness, Accuracy
- Typical External Functions

5 Design Process of Control

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Stability

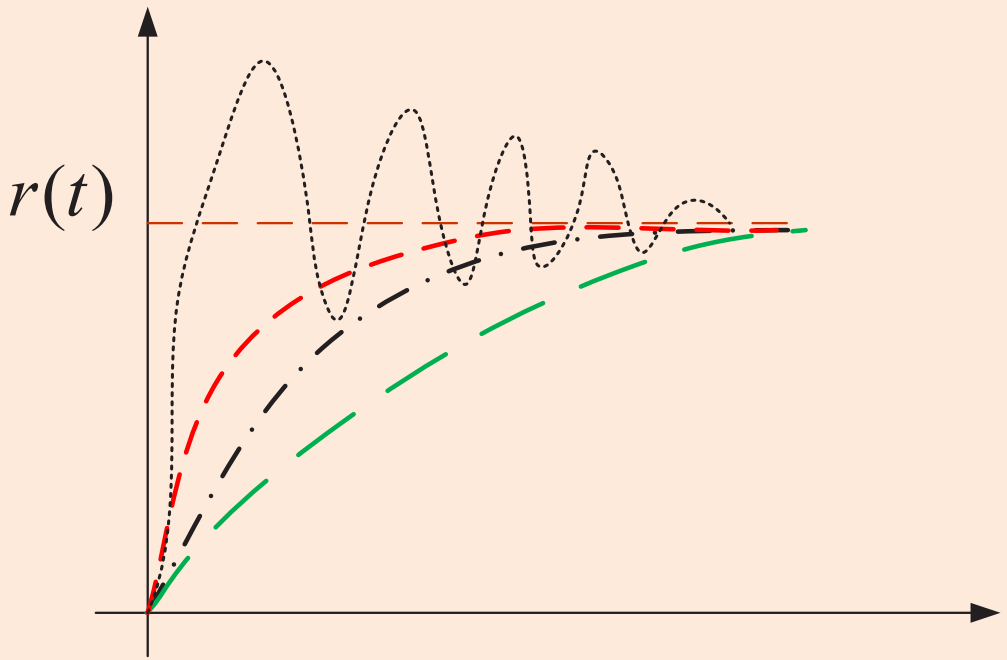
A system is stable, if its output is under control. Otherwise, it is unstable.
A stable system produces a bounded output for a given bounded input. (BIBO)



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Rapidity

The speed for a system to achieve the desired performance.
Good rapidity means short transient process to the desired performance, which having good dynamic property.



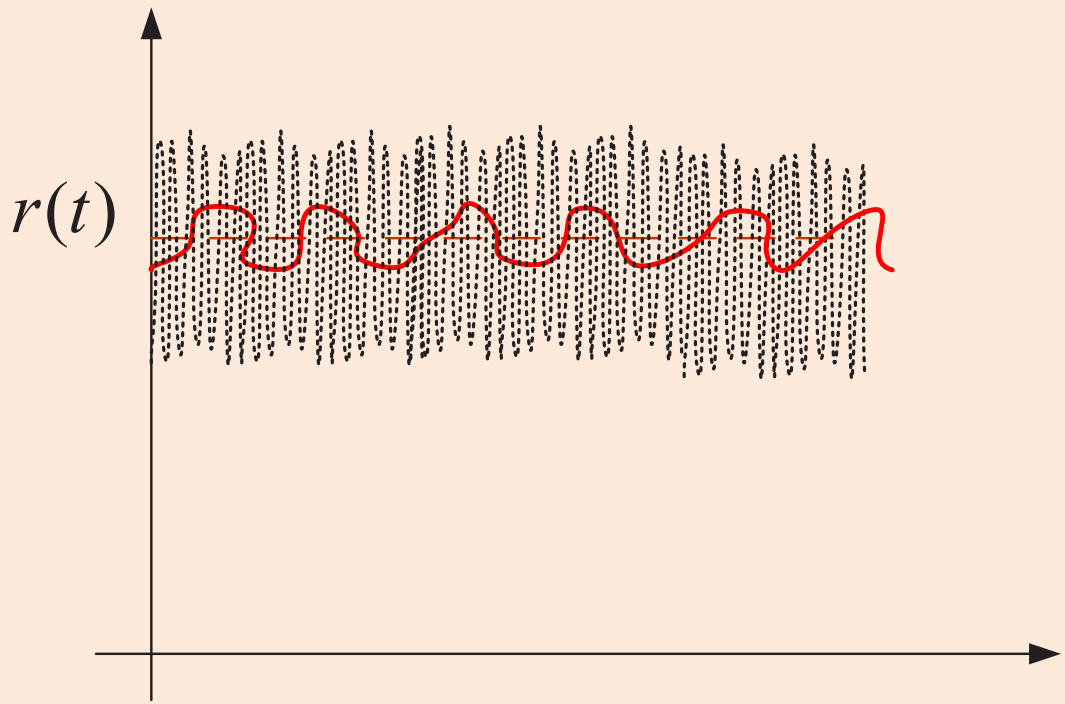
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Smoothness

The degrees of oscillation and vibration. The performance of system output and control input for the desired performances.

Accuracy

The ability of control output approaching to the reference / desired command.



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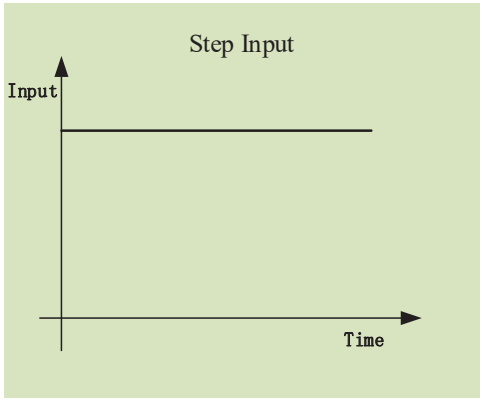
1 History of Automatic Control

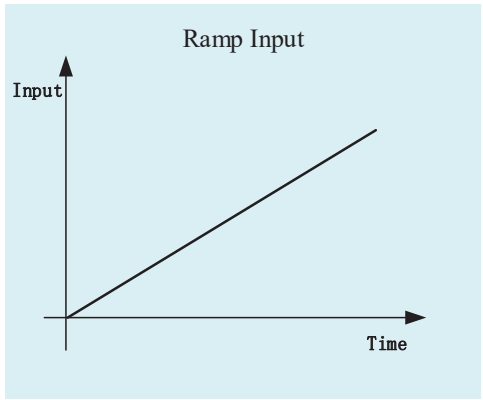
2 Fundamentals of Control

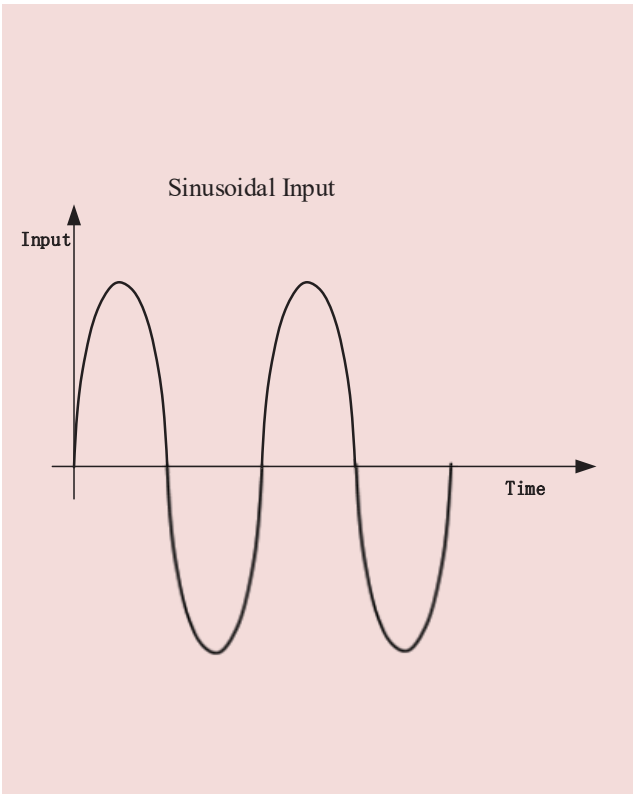
3 Classification of Automatic Control Mode

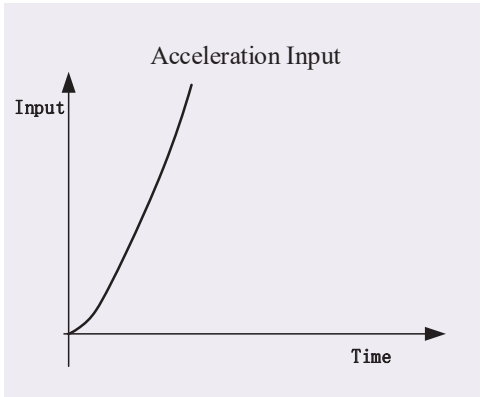
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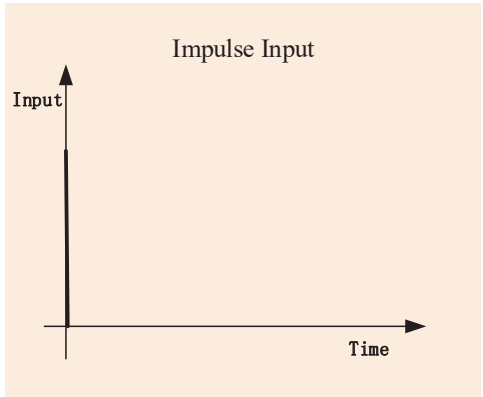
5 Design Process of Control











Series Expand: $r(t) \approx \sum_{i=1}^{n} c_i r_i(t)$

Taylor series, Maclaurin series

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- Typical Process
- Other Control Issues

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Typical Design Process

Step1 Modeling & Identification
Physics based ODE models, Transfer function, State-space model, Data-driven model, $\cdots$

Step2 Analysis
Stability, Root-locus, Controllability $\&$ Observability $\cdots$

Step3 Design
Classical: Lead, Lag, Lead-lag, PID,
Modern: Pole-assignment, LQR...
Robust and Nonlinear control

Step4 Simulation (Alternatively)
Computer language based method: C, C++, M, C $\sharp$, $\cdots$
Block: Simulink, Mathematica, $\cdots$
Vision: Gazebo, Webots, $\cdots$

Step5 Implementation
PLC, Embedded computer, MCU, DSP, FPGA, Analog and Digital Circuits, Mechanical machine...

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- Sensor-fusion, Data-fusion
- Filter, State Estimation
- Parameter Identification
- Prediction
- Planning
- Trajectory Optimization
- Intelligent Learning Control
- Multi-agent
- Simulation, etc...

Problems and Methods might vary, but the basic ideas are Similar.



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